

Chapter 10: Light — Reflection and Refraction

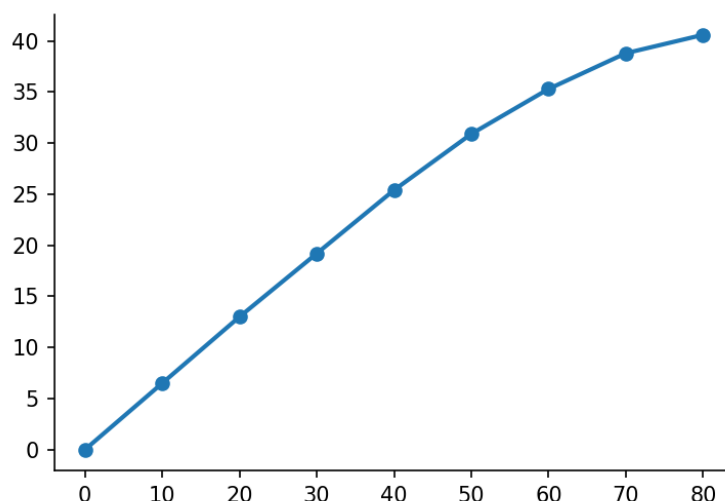
NCERT Class 10 Science — Physics Unit

Light is a form of energy that enables us to see objects around us. When light travels from one transparent medium to another, it changes its speed and, as a result, bends at the interface of the two media. This phenomenon is known as refraction of light. The bending of light also occurs when it strikes a smooth polished surface such as a mirror, and in this case the phenomenon is called reflection. Reflection follows two laws: the angle of incidence is always equal to the angle of reflection, and the incident ray, the reflected ray, and the normal at the point of incidence all lie in the same plane. Refraction, on the other hand, occurs because light travels at different speeds in different media, and it is quantitatively described by Snell's Law, which relates the angle of incidence to the angle of refraction through the refractive indices of the two media involved.

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

where n_1 and n_2 are the refractive indices of medium 1 and medium 2, and θ_1 and θ_2 are the angle of incidence and angle of refraction respectively.

Figure 10.3



The graph above shows how the quantities behave as one of them increases. Observe the trend.

Table 10.1 — Refractive indices of common media

Medium	Refractive Index	Speed of Light (approx.)
Vacuum	1.00	3.00×10^8 m/s
Water	1.33	2.25×10^8 m/s
Glass (crown)	1.52	1.97×10^8 m/s
Diamond	2.42	1.24×10^8 m/s

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

This is the lens formula relating object distance (u), image distance (v), and focal length (f) for a spherical lens, following the sign convention used in this chapter.

Total internal reflection occurs when light travelling from a denser medium to a rarer medium strikes the boundary at an angle greater than the critical angle, causing all the light to be reflected back into the denser medium rather than refracted. This principle is used in optical fibres, prism periscopes, and mirages observed on

hot roads.